

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

Name	Vaishnavi.S
Register Number	192521313

COURSE OUTCOME COVEREDCO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Duration : 60 Minutes

Code of Conduct

I Vaishnavi.S (192521313) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

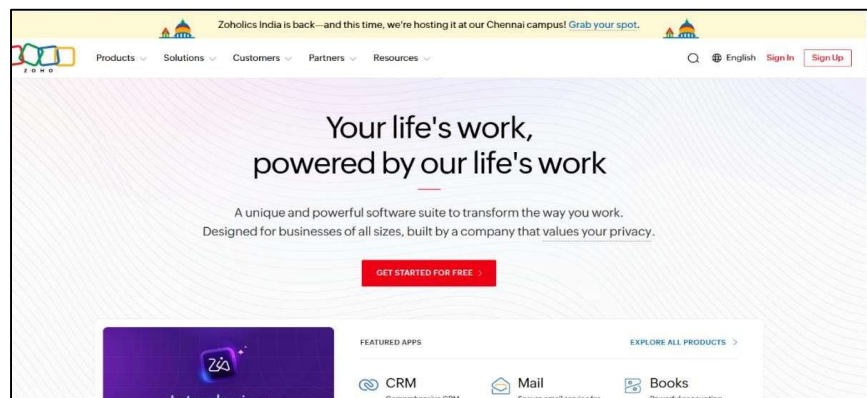
Aim: To create a cloud-based Library Book Reservation System using Zoho Creator.

Steps:

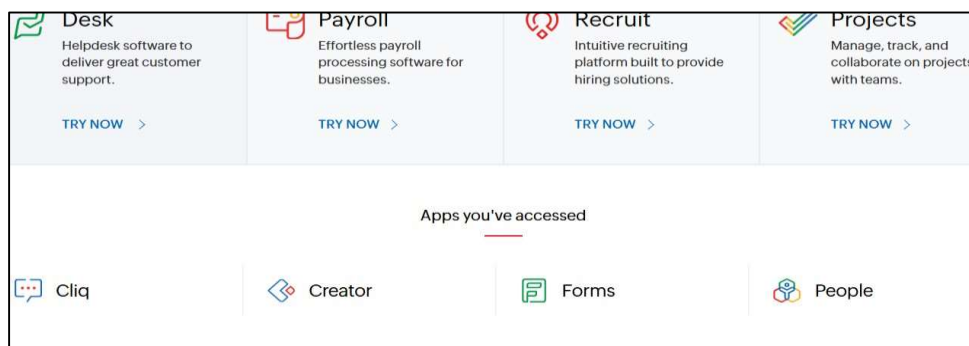
1. Create a new application.
2. Create a Book Reservation form.
3. Add fields: Book Title, Author, Publication, Edition, Rack Number, Number of Copies, and Status.
4. Create a report to display available and reserved books.
5. Test the application with sample book records.

Implemented

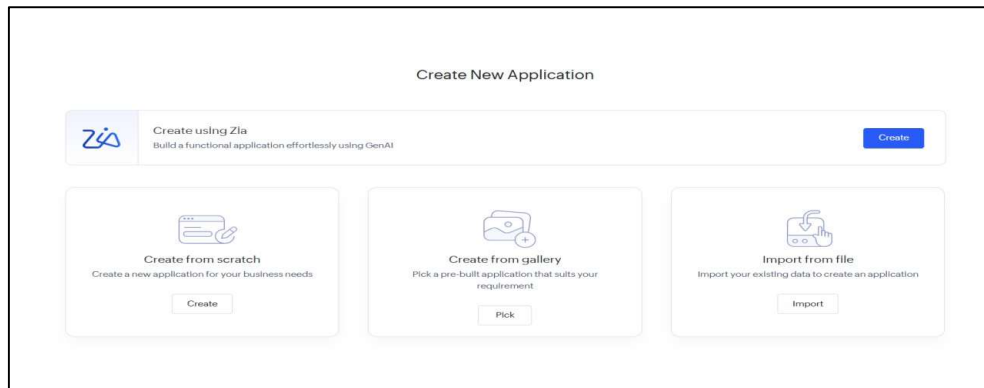
STEP 1: Go to zoho.com



STEP 2: Login to your Zoho account and open Zoho Creator



STEP 3: Click Create New Application and select Create from Scratch



STEP 4: Enter Library Book Reservation System as the application name and click Create Application.

The screenshot shows the 'Create from scratch' form. It has a title 'Create from scratch' with a logo above it. Below the title, there's a form with the following fields: 'Application Name' (with a red asterisk) containing 'Library Book Reservation System', and a checkbox 'Enable Environment' with a 'Learn More' link. At the bottom, there's a large blue 'Create Application' button.

Step 5: Add fields: Book Title, Author, Publication, Edition, Rack Number, Number of Copies, and Status

The screenshot shows the 'Library Management System' application interface. On the left, there's a sidebar with a menu: 'Dashboard', 'Issue Records' (selected), 'Issue Records Rep...', 'Board', 'Borrowing History', 'Books', 'Students', 'Return Records', 'Dashboard Metrics', and 'Items'. The main area is titled 'Issue Records' and contains a form with the following fields: 'Student' (text input), 'Book' (dropdown menu), 'Issue Date' (date input), 'Return Date' (date input), 'Due Date' (date input), 'Status' (dropdown menu), 'Fine Amount' (text input), and 'Returned By' (text input). At the bottom, there are 'Submit' and 'Reset' buttons.

SIMATS Library Management Sys...

Book Issue Return

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123

Number

Date

Time

Drop Down

Transaction Date

Issue Date

Due Date

Return Date

Fine Amount

Transaction Status

Transaction Code

Reservation

Student

Book

SIMATS Library Management System

SIMATS Library Manageme...

Book Details

Dashboard

Book Issue Return

Book Reservations

Book Details

+ Book Details

Book Details Report

Book Categories

Student Registrati...

Fine Payments

Book Title

Author

Publication

Publication Year

Edition

ISBN Number

Book Category

Number of Copies

Available Copies

Rock Number

Shelf Number

SIMATS Library Manageme...

Student Registration

Dashboard

Book Issue Return

Book Reservations

Book Details

Student Registrati...

+ Student Registration

All Student Registr...

Fine Payments

Student Name

First Name

Department

Course

Year

Section

Mobile Number

Email Address

Address

Address Line 1

Registration Date

dd-MMM-yyyy

Student Code

STEP 6: Save the form, enter sample booking records, and view them in the Library Book Reservation System.

Transaction Date	Issue Date	Due Date	Return Date	Fine Amount	Transaction Status	Transaction Code	Reservation
11-Aug-2026	11-Aug-2026	18-Aug-2026	18-Aug-2026	\$ 0.00	Returned	TRX005	RES-2026-002
10-Aug-2026	10-Aug-2026	17-Aug-2026	17-Aug-2026	\$ 0.00	Taken	TRX004	RES-2026-001
09-Aug-2026	09-Aug-2026	16-Aug-2026	16-Aug-2026	\$ 0.00	Taken	TRX003	RES-2026-001
08-Aug-2026	08-Aug-2026	15-Aug-2026	15-Aug-2026	\$ 0.00	Taken	TRX002	RES-2026-004
07-Aug-2026	07-Aug-2026	14-Aug-2026	14-Aug-2026	\$ 0.00	Returned	TRX001	RES-2026-004

SIMATS Library Manageme...

Dashboard

Book Issue Return

Book Reservations

Book Details

Student Registrati...

Student Registration

All Student Registr...

Fine Payments

All Student Registrations

Student Name

Department

Course

Year

Section

Mobile Number

Email Address

Address

Register

Robert Chen

Engineering

Software Design

2026

B

+19015986438

robertc@university.edu

505 Eng Blvd

07-Aug

Maria Garcia

Biology

Cell Biology

2026

C

+18729908100

mariag@university.edu

404 Bio Way

07-Aug

James Wilson

Mathematics

Calculus III

2026

A

+18298545159

jamesw@university.edu

303 Math Ln

07-Aug

Sarah Connor

Physics

Quantum Mechanics

2026

B

+18192467340

sarahc@university.edu

202 Science Rd

07-Aug

Alex Morgan

Computer Science

Data Structures

2026

A

+17022597323

alexm@university.edu

101 Campus Dr

07-Aug

All Fine Payments

Payment Date

Payment Method

Payment Status

Payment Code

Student

Transaction

Fine Amount

07-Aug-2026

Cash

Paid

PAY-005

STU2026004

TRX001

\$ 75.00

07-Aug-2026

Credit Card

Pending

PAY-004

STU2026003

TRX002

\$ 10.00

07-Aug-2026

Debit Card

Paid

PAY-003

STU2026002

TRX004

\$ 50.25

07-Aug-2026

Credit Card

Paid

PAY-002

STU2026005

TRX004

\$ 15.00

07-Aug-2026

Credit Card

Paid

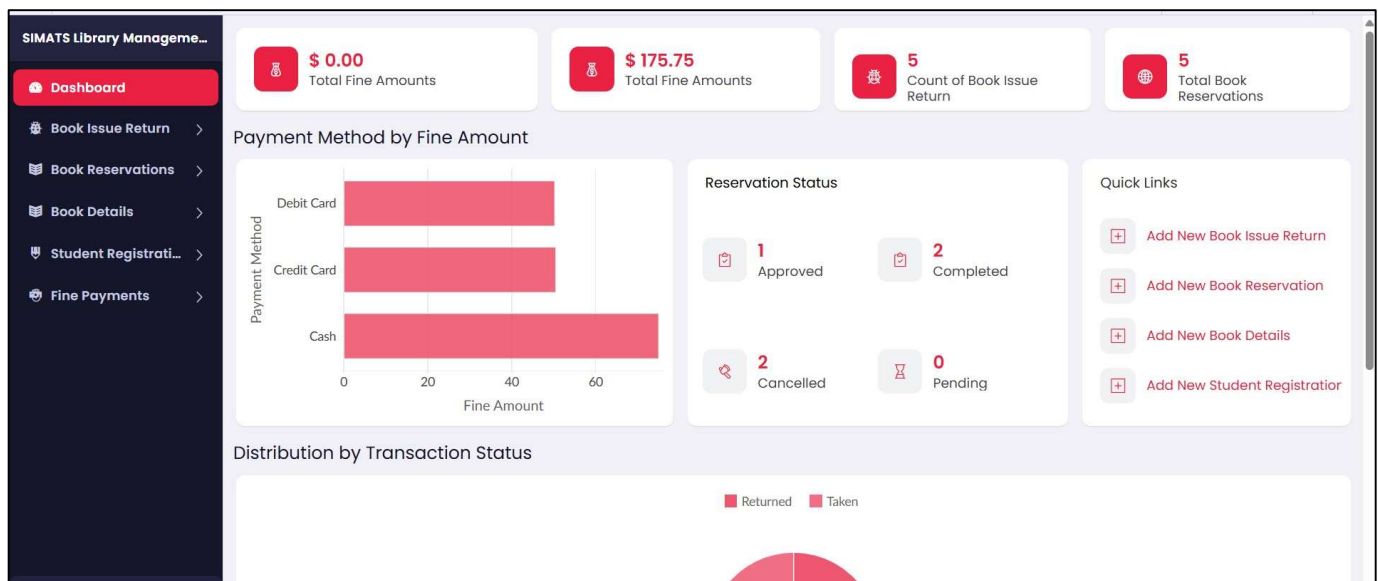
PAY-001

STU2026001

TRX005

\$ 25.50

STEP 7: Final Dashboard.



Result: Successfully created a cloud-based Library Book Reservation System using Zoho Creator.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org, over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

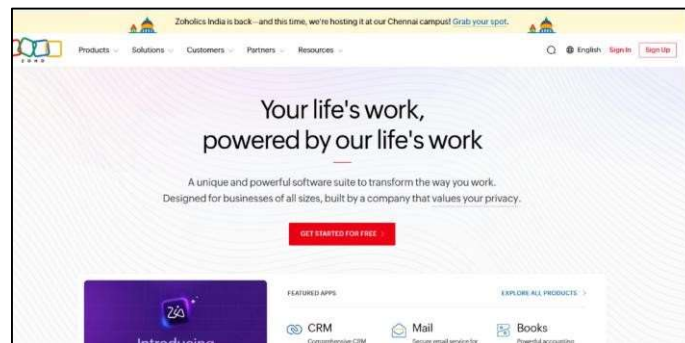
Aim: To create a cloud-based Payroll Processing System using Zoho Creator.

Steps:

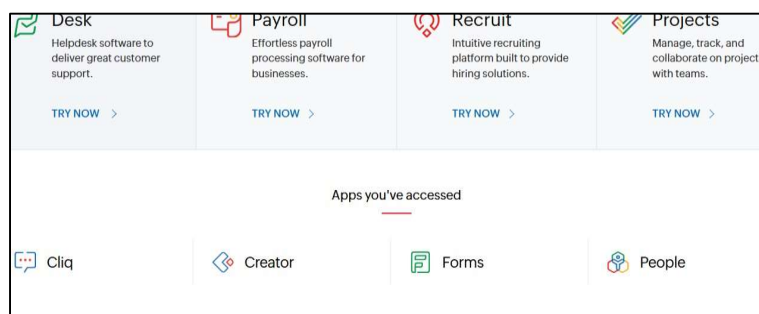
1. Create a new application.
2. Create an Employee Payroll form.
3. Add fields: Employee Name, Experience, Basic Pay, HRA, DA, TA, Gross Pay, and Net Pay.
4. Use formulas to calculate Gross Pay and Net Pay.
5. Save employee details and verify the payroll report.

Implemented:

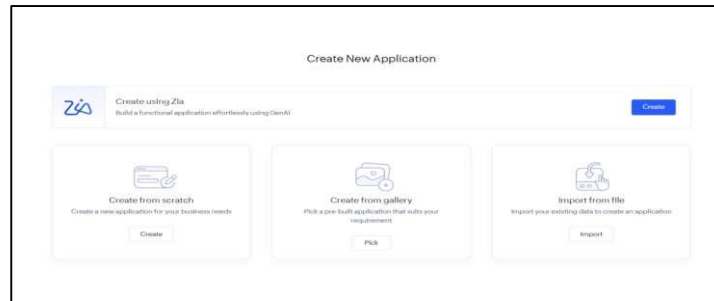
Step 1: Go to zoho.com



STEP 2: Login to your Zoho account and open Zoho Creator



STEP 3: Click Create New Application and select Create from Scratch



STEP 4: Enter Payroll Processing System as the application name and click Create Application.

The screenshot shows the 'Create from scratch' form. At the top, there's a header 'Create from scratch'. Below it, there's a text input field for 'Application Name' with the value 'Payroll Processing System'. There is a checkbox for 'Enable Environment' and a 'Learn More' link. A large blue button labeled 'Create Application' is at the bottom.

Step 5: Add fields: Employee Name, Employee Code, Salary, Total, Average, Percentage, and PaySlip.

The screenshot shows the 'Employee Registration' form. It has a sidebar with navigation links: Dashboard, Employee Registr..., Employee Registra..., All Employee Regi..., Genders, Salary Structures, Monthly Payrolls, Payslip, Management, Zoho People Depa..., and Zoho People Desig... The main form has fields for: Employee Name (highlighted), Employee Code, Gender (dropdown), Date of Birth (dd-MM-yyyy), Department (dropdown), Designation (dropdown), Date of Joining (dd-MM-yyyy), Experience in Present Org (text), Overall Experience (text), and Mobile Number (with a country code dropdown and a text field containing 81234 56789). There is also an 'Email Address' field at the bottom.

The screenshot shows the 'Salary Structure' form. It has a sidebar with navigation links: Dashboard, Employee Registr..., Salary Structures (highlighted), All Salary Structures, Monthly Payrolls, Payslip, Management, Zoho People Depa..., and Zoho People Desig... The main form has fields for: Employee (highlighted), Basic Pay, House Rent Allowance, Dearness Allowance, Travel Allowance, Medical Allowance, Special Allowance, Bonus, Provident Fund, and Professional Tax.

Cloud Payroll Processing S...

Dashboard

Employee Registr...

Salary Structures

Monthly Payrolls

Monthly Payroll

All Monthly Payrolls

Statuses

Payslip

Management

Zoho People Depa...

Zoho People Desig...

Monthly Payroll

Payroll Month

Payroll Year

Employee

Working Days

Days Present

Leave Days

Overtime Hours

Overtime Amount

Gross Pay

Total Deductions

Net pay

STEP 6: Save the form Payroll Processing, enter sample booking records, and view them in the System.

Cloud Payroll Processing S...

Dashboard

Employee Registr...

Salary Structures

Monthly Payrolls

Monthly Payroll

All Monthly Payrolls

Statuses

Payslip

Management

Zoho People Depa...

Zoho People Desig...

All Monthly Payrolls

Payroll ID	Payroll Month	Payroll Year	Employee	Working Days	Days Present	Leave Days	Overtime Hours	Overtime A
5	April	2026	Robert Williams Sarah Martinez Michael Chen Jennifer Taylor	22	18	4	6	\$
4	May	2026	Alice Johnson	22	22	0	4	\$
3	June	2026	Alice Johnson Robert Williams Michael Chen	22	18	3	6	\$
2	July	2026	Alice Johnson	22	21	1	5	\$
1	August	2026	Alice Johnson Robert Williams Sarah Martinez	22	20	2	4	\$

Cloud Payroll Processing S...

Dashboard

Employee Registr...

Salary Structures

Monthly Payrolls

Monthly Payroll

All Monthly Payrolls

Statuses

Payslip

Payslip Report

Management

Zoho People Depa...

Zoho People Desig...

Payslip Report

Employee Name	Employee Code	Department	Designation	Payroll Month	Payroll Year	Basic Pay	HRA	DA
Amit Verma	EMP005	Ops	Operations Head	Aug	2026	\$ 60,000.00	\$ 12,000.00	\$ 8,000.00
Priya Patel	EMP004	Sales	Sales Executive	Aug	2026	\$ 35,000.00	\$ 7,000.00	\$ 3,500.00
Vikram Singh	EMP003	Finance	Accountant	Aug	2026	\$ 40,000.00	\$ 8,000.00	\$ 4,000.00
Anita Sharma	EMP002	HR	HR Manager	Aug	2026	\$ 45,000.00	\$ 9,000.00	\$ 4,500.00
Rajesh Kumar	EMP001	IT	Senior Developer	Aug	2026	\$ 50,000.00	\$ 10,000.00	\$ 5,000.00

Cloud Payroll Processing S...

Dashboard

Employee Registr...

Salary Structures

Monthly Payrolls

Monthly Payroll

All Monthly Payrolls

Statuses

Payslip

Management

Leave Management

All Management

Leave Types

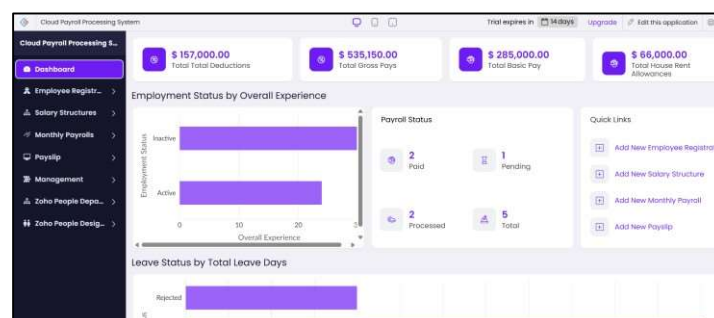
Zoho People Depa...

Zoho People Desig...

All Management

Employee	Leave Type	From Date	To Date	Total Leave Days	Leave Status
Alice Johnson Robert Williams Sarah Martinez Michael Chen Jennifer Taylor	Unpaid Leave	21-Aug-2026	25-Aug-2026	4	Pending
Alice Johnson Robert Williams	Sick Leave	16-Aug-2026	20-Aug-2026	5	Approved
Alice Johnson Robert Williams	Unpaid Leave	11-Aug-2026	15-Aug-2026	4	Rejected
Alice Johnson Robert Williams Sarah Martinez	Sick Leave	06-Aug-2026	10-Aug-2026	5	Pending
Alice Johnson Robert Williams Sarah Martinez Michael Chen Jennifer Taylor	Casual Leave	01-Aug-2026	05-Aug-2026	3	Pending

STEP 7: Final Dashboard



Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Aim: To create a virtual machine with 1 cpu, 2gb ram and 15gb storage disk using a type 2 virtualization software.

Procedure:

STEP 1: Download VMware workstation and installed as type 2hypervisor.

STEP 2: Download ubuntu or tiny OS as iso image file.

STEP 3: In VMware workstation->create new VM.

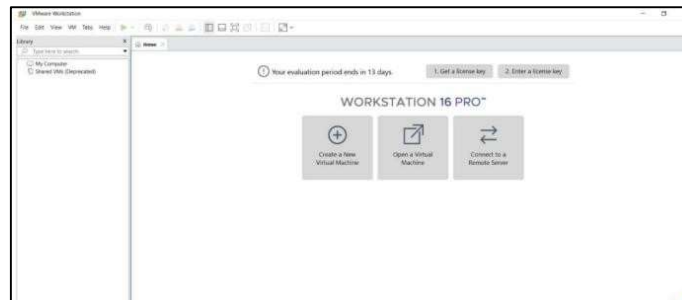
STEP 4: Do the basic configuration settings.

STEP 5: Created tiny OS virtual machine.

STEP 6: Launch the VM.

Implemenation:

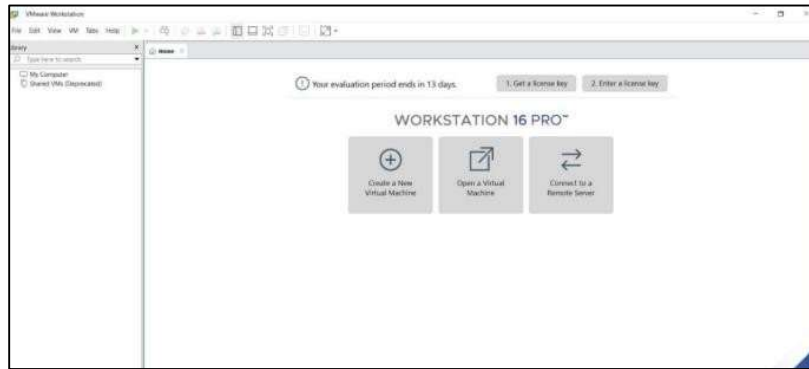
STEP 1: DOWNLOAD VMWARE WORKSTATIONAND INSTALLED AS TYPE 2 HYPERVISOR



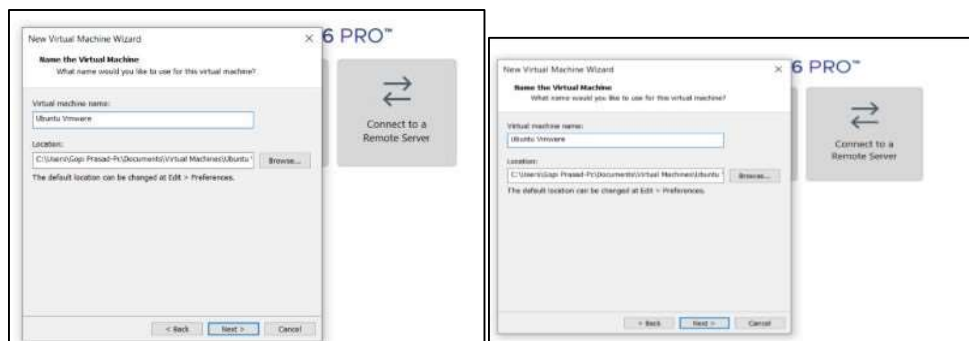
STEP2: DOWLOAD UBUNTU OR TINY OS AS ISO IMAGE FILE

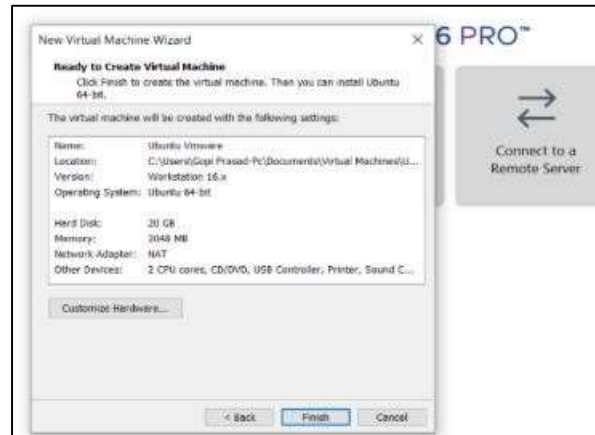
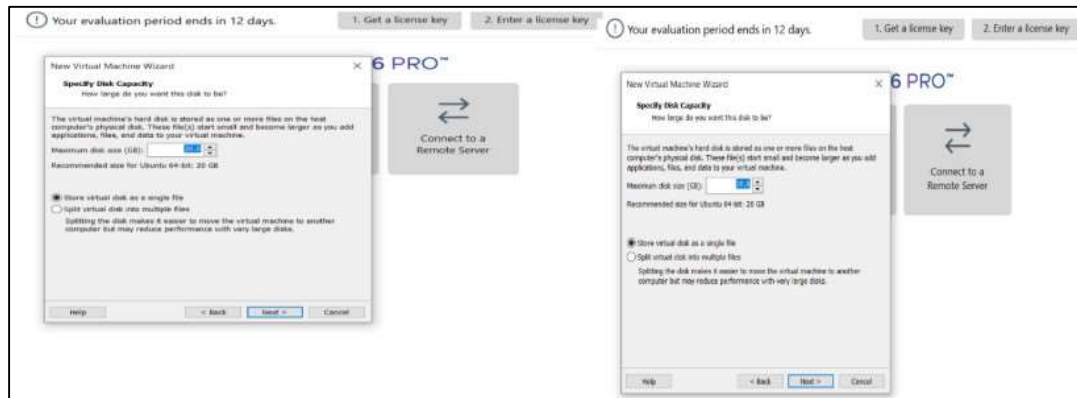
Index of /11.x/x86/release/		
..		
distribution_files/	09-Feb-2020 11:50	-
scsi/	03-Dec-2019 11:14	-
Core-11.1.iso	01-Apr-2020 07:49	14757888
Core-11.1.iso.md5.txt	01-Apr-2020 07:49	48
Core-11.1.iso.xsyzm	01-Apr-2020 07:49	50639
Core-current.iso	01-Apr-2020 07:49	14757888
CorePlus-11.1.iso	01-Apr-2020 07:50	216006056
CorePlus-11.1.iso.md5.txt	01-Apr-2020 07:50	52
CorePlus-11.1.iso.xsyzm	01-Apr-2020 07:50	369358
CorePlus-current.iso	01-Apr-2020 07:50	216006056
TinyCore-11.1.iso	01-Apr-2020 07:50	19922944
TinyCore-11.1.iso.md5.txt	01-Apr-2020 07:50	52
TinyCore-11.1.iso.xsyzm	01-Apr-2020 07:50	68591
TinyCore-current.iso	01-Apr-2020 07:50	19922944

STEP 3: IN VMWARE WORKSTATION->CREATE NEW VM

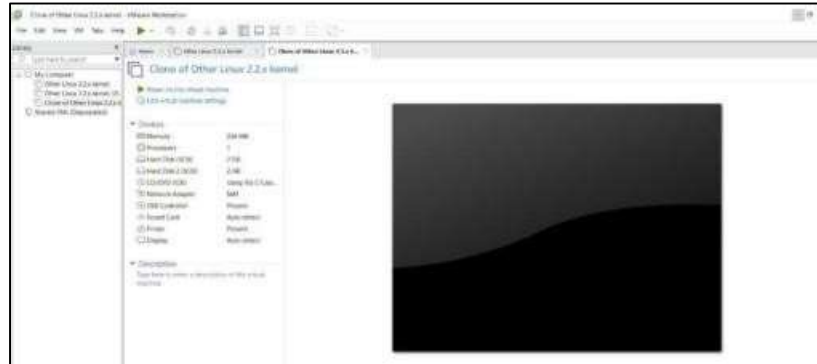


STEP 4: DO THE BASIC CONFIGURATION SETTINGS.





STEP 5: CREATED TINYOS VIRTUAL MACHINE



Result: Successfully created a virtual machine with 1 cpu, 2gb ram and 15gb storage disk using a type 2 virtualization software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Aim: To create a cloning of a vm and test it by loading the previous version/cloned vm.

Procedure:

STEP 1: GO TO VM AND GOTO MANAGE AND CLICK CLONE

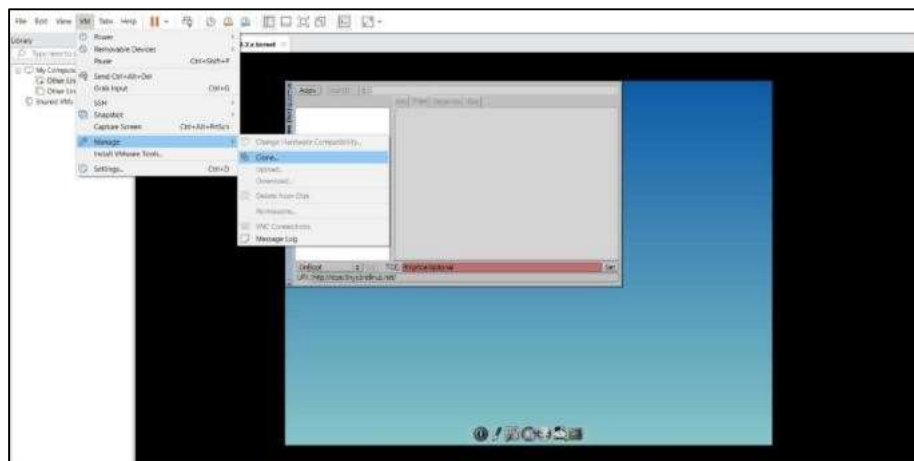
STEP 2: CLICK CLONE

STEP 3: SELECT THE FULL CLONE

STEP 4: AFTER CLONE AGAIN OR VM IS OPENED.

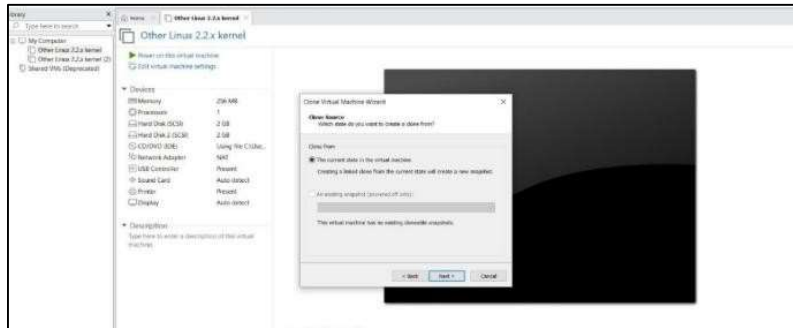
Implementation: CLONING OF A VM

STEP 1: GO TO VM AND GOTO MANAGE AND CLICK CLONE

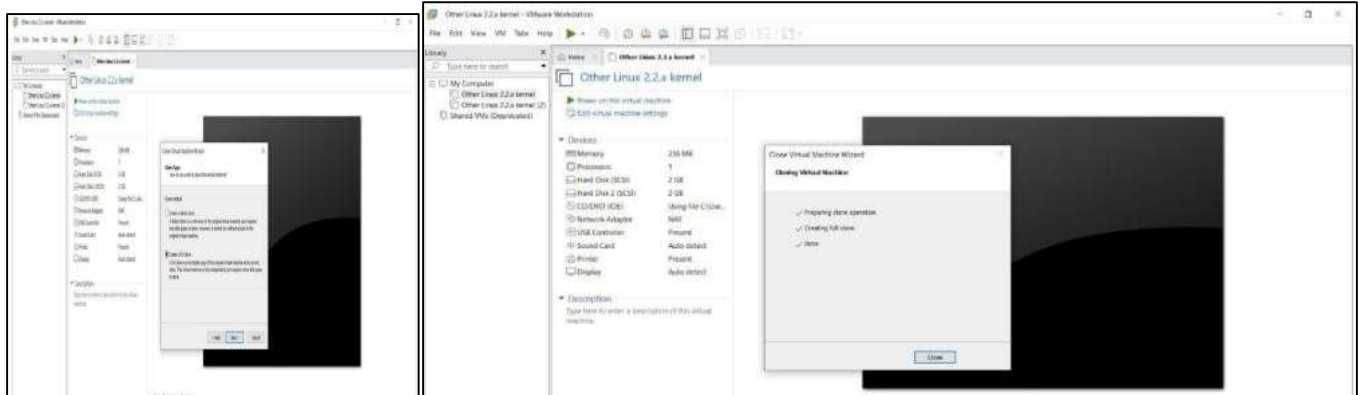


STEP 2: CLICK CLONE





STEP 3: SELECT THE FULL CLONE



STEP 4: AFTER CLONE AGAIN OR VM IS OPENED



Result: Successfully created a cloning of a vm and test it by loading the previous version/cloned vm.